

Shinhwan Kang

✉ shinhwan.kang@kaist.ac.kr | 🌐 github.com/ShinhwanKang | 🔗 linkedin.com/in/shinhwankang | 🐦 @Shinhwan_Kang
| 🎓 Google Scholar



Research Interest [Related Publications]

Data Augmentation [C7], Large Language Models [C6-9], Dataset Construction [C5, J2], Healthcare/Bioinformatics [C7, J4], Graph Summarization Algorithm [C1,C2]

Education

Korea Advanced Institute of Science and Technology (KAIST)

Ph.D. Candidate in Kim Jaechul Graduate School of AI (Advisor - Kijung Shin)

Seoul, Republic of Korea

Mar. 2023 - Current

Korea Advanced Institute of Science and Technology (KAIST)

M.S. Student in Kim Jaechul Graduate School of AI (Advisor - Kijung Shin)

Seoul, Republic of Korea

Mar. 2021 - Feb. 2023

Korea University

Bachelor of Engineering in Computer Science and Engineering

- Total GPA of 3.99/4.5 (94.2/100), Major GPA of 4.09/4.5
- Honor student (17-Spring/Fall, 19-Fall)
- Highest honor student (18-Spring/Fall, 19-Spring, 20-Spring/Fall)

Seoul, Republic of Korea

Mar. 2017 - Feb. 2021

Publications

[C9] Zero-Shot Tabular Learning via Hypergraph Reasoning with LLMs

EMNLP 2026 (Conference on Empirical Methods in Natural Language Processing)

Geon Lee, Kyungho Kim, Dongwon Choi, Juyeon Kim, **Shinhwan Kang**, and Kijung Shin

[C8] Training-Free LLM-Based Recommendation with Post-LLM Item Refinement Using Collaborative Signals

CIKM 2026 (Short Paper) (ACM Conference on Information and Knowledge Management 2026) [Paper | Github]

Kyungho Kim, Sunwoo Kim, Geon Lee, **Shinhwan Kang**, Sojeong Kim, Liam Collins, Bhuvesh Kumar, Donald Loveland, and Kijung Shin

[C7] Improving Rare Medication Recommendation with Counterfactual Data Augmentation and Large Language Models

RecSys 2026 (ACM Conference on Recommender Systems) (Acceptance Ratio $\approx 18\%$) [Paper | Github]

Shinhwan Kang, Soo Yong Lee, Jaewon Kim, Kijung Shin, and Buru Chang

[C6] ReFuGe: Feature Generation for Prediction Tasks on Relational Databases with LLM Agents

WWW 2026 (Short Paper) (ACM International World Wide Web Conference) [Paper | Github]

Kyungho Kim, Geon Lee, Juyeon Kim, Dongwon Choi, **Shinhwan Kang**, and Kijung Shin

[C5] RDB2G-Bench: A Comprehensive Benchmark for Automatic Graph Modeling of Relational Databases

NeurIPS 2025 (Advances in Neural Information Processing Systems) [Paper | Github | Document | Dataset]

Dongwon Choi, Sunwoo Kim, Juyeon Kim, Kyungho Kim, Geon Lee, **Shinhwan Kang**, and Kijung Shin

[J4] Gene expression inference based on graph neural networks using L1000 data

Briefings in Bioinformatics [Paper | Github]

Tae Hyun Kim*, Harim Kim*, Hyunjin Hwang*, **Shinhwan Kang***, Kijung Shin, and Inwha Baek

[J3] Inductive Influence Estimation and Maximization over Unseen Social Networks under Two Diffusion Models

Data Mining and Knowledge Discovery [[Paper](#) | [Github](#)]

Jihoon Ko*, Sojeong Kim*, Kyuhan Lee, **Shinhwan Kang**, Dongyeong Hwang, Kijung Shin, and Noseong Park

[J2] BeGIn: Extensive Benchmark Scenarios and An Easy-to-use Framework for Graph Continual Learning

ACM TIST (Transactions on Intelligent Systems and Technology) [[Paper](#) | [Github](#) | [Document](#)]

Jihoon Ko*, **Shinhwan Kang***, Taehyung Kwon, Heechan Moon, and Kijung Shin

[C4] Rethinking reconstruction-based graph-level anomaly detection: limitations and a simple remedy

NeurIPS 2024 (Advances in Neural Information Processing Systems) [[Paper](#) | [Github](#) | [Poster](#)]

Sunwoo Kim, Soo Yong Lee, Fanchen Bu, **Shinhwan Kang**, Kyungho Kim, Jaemin Yoo, and Kijung Shin

[C3] Hypeboy: Generative self-supervised representation learning on hypergraphs

ICLR 2024 (International Conference on Learning Representations) [[Paper](#) | [Github](#) | [Poster](#)]

Sunwoo Kim, **Shinhwan Kang**, Fanchen Bu, Soo Yong Lee, Jaemin Yoo, and Kijung Shin

[J1] Interplay between Topology and Edge Weights in Real-World Graphs: Concepts, Patterns, and an Algorithm

Data Mining and Knowledge Discovery [[Paper](#) | [Github](#) | [Poster](#)]

Fanchen Bu, **Shinhwan Kang**, and Kijung Shin

[W1] Region-Conditioned Orthogonal 3D U-Net for Weather4Cast Competition

NeurIPS 2022 (Workshop Competition Track) (Advances in Neural Information Processing Systems) [[Paper](#) | [Github](#)]

Taehyeon Kim, **Shinhwan Kang**, Hyeonjeong Shin, Deukryeol Yoon, Seongha Eom, Kijung Shin, and Se-Young Yun

[C2] Personalized Graph Summarization: Formulation, Scalable Algorithms, and Applications

ICDE 2022 (International Conference on Data Engineering) (Acceptance Ratio $\approx 27\%$) [[Paper](#) | [Github Appendix](#) | [Video](#) | [Slides](#)]

Shinhwan Kang, Kyuhan Lee, and Kijung Shin

[C1] Are Edge Weights in Summary Graphs Useful? - A Comparative Study

PAKDD 2022 (Pacific-Asia Conference on Knowledge Discovery and Data Mining) (Acceptance Ratio $\approx 19.3\%$) [[Paper](#) | [Github](#) | [Appendix](#) | [Slides](#)]

Shinhwan Kang, Kyuhan Lee, and Kijung Shin

Projects

EntireDB2AI

Software Technology Advanced Research Lab (StarLab)

- Proposed a LLM-based relational database (RDB) augmentation method.

Republic of Korea

Mar. 2025 - Current

Deep Learning-Based Precipitation Nowcasting

National Institute of Meteorological Sciences

- Developed a classification-based precipitation nowcasting model for short-term forecasts (up to 6 hours, in 10-minute intervals)
- Processed and utilized terabyte (TB)-scale radar and satellite datasets for training and evaluation

Republic of Korea

Mar. 2021 - Feb. 2025

Experience

ML X, NAVER

Internship

- Contributed to the development of a conversational recommender system powered by HyperClova
- Explored dataset construction techniques for HyperClova fine-tuning, utilizing large language models

Seongnam, Republic of Korea

Mar. 2023 - Jun. 2023

KAIST Data Mining Lab, KAIST

Individual Researcher (Advisor - Kijung Shin)

- Participated in individual project on graph summarization algorithms

Daejeon, Republic of Korea

Jul. 2020 - Feb. 2021

DMIS (Data Mining & Information System) Laboratory, Korea University

Undergraduate Researcher (Advisor - Jaewoo Kang)

- Participated in a lead optimization project and many research efforts in machine learning, especially deep learning

Seoul, Republic of Korea

Mar. 2018 - Jun. 2020

Brain Engineering Laboratory, Korea University

Undergraduate Researcher (Advisor - Sunhee Kim)

Seoul, Republic of Korea

Feb. 2019 - Jul. 2019

- Participated in projects on brain signal data collection and processing
- Developed real-time intelligent emotion recognition technology based on multi-modal brain signals

Extra Curricular Activities

Microsoft Student Partner Developer

Republic of Korea

Microsoft

Jul. 2017 - Aug. 2018

- Recommendation system with Azure Machine Learning (Nov. 2017, Dec. 2017)
- Continuous distribution of Web App using GitHub (Jun. 2018)
- Azure Functions & Cognitive API (Jul. 2018)
- Attending the Asia-Pacific Microsoft Students Partners Summit at the National University of Singapore (NUS) in Singapore on 2018.05.18-20.

Award/Recognition

Excellence Award Competition (3th)

Changwon Exhibition Convention Center, South

Korea

National Institute of Meteorological Sciences

Nov. 2025

- Developed a classification-based precipitation nowcasting model for short-term forecasts (up to 6 hours, in 10-minute intervals)
- Processed and utilized terabyte (TB)-scale radar and satellite datasets for training and evaluation

Weather4Cast Competition (4th)

Online

NeurIPS

Dec. 2022

- Predict super-resolution rain movies
- Transfer Learning across space and time under strong shifts
- Exploit data fusion to model graph radar and multi-band satellite images

Project for Graduation Competition (1st)

Online

Korea Univ. Computer Science and Engineering

Jun. 2020

- Lead Optimization utilizing Gene Expression Profile [\[Leader Board | Video\]](#)

AI Dataset Challenge (3rd)

Online

SELECT STAR

Apr. 2020

- Propose "Korean Blank Reasoning Dataset" Based on Crowdsourcing

Academic Services

Reviewer@CIKM 2026 (Full, Applied Research Track)

Online

ACM International Conference on Information and Knowledge Management

Reviewer@SIGIR 2026

Online

ACM Special Interest Group on Information Retrieval

Reviewer@WWW 2026

Online

ACM International World Wide Web Conference

Reviewer@CIKM 2025 (Short Track)

Online

ACM International Conference on Information and Knowledge Management

Reviewer@NeurIPS 2025, 2026

Online

Advances in Neural Information Processing Systems

Teaching Assistant@Upstage

Boostcamp AI Tech, Upstage

Machine Learning with Graphs

Dec. 2020 - Mar. 2021

Skills/Languages

Programming Python, Java, C

Languages Korean (Mother Tongue), English (Fluent)